

MID TERM EXAMINATIONB.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — 3rd SEMESTER, JANUARY 2023**Paper Code: ES-201****Subject: Computational Methods****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Marks	CO										
1(a)	Find a root of the equation $x^2 - 3x + 2 = 0$ using secant method up to second iteration.	[2.5]	CO1										
1(b)	Write the statement of mean value theorem.	[2.5]	CO1										
1(c)	Find the Roundoff error in storing the number 752.6835 using the four digit mantissa.	[2.5]	CO1										
1(d)	Use Simpson's ■ rule to estimate the integral $\int_1^2 [dx / (1 + x^2)]$.	[2.5]	CO2										
2(a)	Perform four iterations of the Newton-Raphson method to obtain the approximate value of (14)^{1/2} starting with the initial approximation $x_0 = 4$.	[5]	CO1										
2(b)	Minimize $f(x) = x^2$ over (-5, 15) , $n = 5$ by Fibonacci method.	[5]	CO1										
3(a)	For the data: <table><tr><td>X</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>F(x)</td><td>5</td><td>1</td><td>1</td><td>11</td></tr></table> Find the estimated value of f(1.5) .	X	-1	0	1	2	F(x)	5	1	1	11	[5]	CO2
X	-1	0	1	2									
F(x)	5	1	1	11									
3(b)	Using Lagrange interpolation, calculate the profit in the year 2000 from the given table: <table><tr><td>Year</td><td>1997</td><td>1999</td><td>2001</td><td>2002</td></tr><tr><td>Profit (lakh ■)</td><td>43</td><td>65</td><td>159</td><td>248</td></tr></table>	Year	1997	1999	2001	2002	Profit (lakh ■)	43	65	159	248	[5]	CO2
Year	1997	1999	2001	2002									
Profit (lakh ■)	43	65	159	248									
4(a)	Using the Trapezoidal rule, Evaluate $\int_0^3 (x^2 + 1) dx$ by dividing the range into 6 equal intervals. Hence compare the result with exact value.	[5]	CO1										
4(b)	Find the value of $\Delta^4 y_0$ and $\Delta^5 y_0$ for $y_0 = 3, y_1 = 12, y_2 = 81, y_3 = 200, y_4 = 100$.	[5]	CO2										

**** End of Question Paper ****